

PROPHET SGS Quadrature Note

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May 14 2026

1 Current Implementation

<https://www.overleaf.com/project/62979703554e638f067b6c38>

1.1 Cloud Fraction and Microphysics

Cloud fraction and microphysical process rates in subdomain m are evaluated over the intra-subdomain SGS distribution, whose mean is supplied by the prognostic state $(h_s^{(m)}, q_t^{(m)}, q_l^{(m)}, q_i^{(m)})$ and whose spread in (h_s, q_t) is supplied by the covariance $\Sigma_{h_s, q_t}^{\text{iso}}$ assembled from the variance closure (??) and the cross-covariance closure (??). To keep cloud fraction and microphysics mutually consistent and consistent with the prognostic condensate budget, we drive both closures from a single Gauss-Hermite quadrature pass over the SGS PDF that caches the moments of the saturation variable and of the equilibrium cloud condensate. Cloud fraction is then evaluated through a closed-form expression in the coordinate of the saturation variable, while microphysical tendencies are evaluated through a shape-function reconstruction of the local condensate at each quadrature point that conserves the prognostic mean by construction. The prognostic equations carry non-equilibrium condensate (e.g., supersaturation that has not yet condensed at the resolved scale, or persistent supercooled liquid), while the unresolved fluctuations within each subdomain are assumed to relax rapidly on the saturation curve [?, ?, ?].

Shared SGS moments pre-pass. We sample the joint $(\delta h_s, \delta q_t)$ SGS PDF on a 2-D Gauss-Hermite grid via a mapping $\vec{g}^{(m)} : \mathbb{R}^2 \rightarrow (h_s, q_t)$ that takes a standard Gaussian latent variable $\vec{\xi} \sim \mathcal{N}(\vec{0}, \mathbf{I})$ to the chosen distribution centered on subdomain m . For Gaussian (h_s, q_t) , $\vec{g}^{(m)}$ is the affine map $\vec{g}^{(m)}(\vec{\xi}) = (h_s^{(m)}, q_t^{(m)}) + \mathbf{L} \vec{\xi}$ with $\mathbf{L}$ the Cholesky factor of $\Sigma_{h_s, q_t}^{\text{iso}}$; for lognormal q_t , $(h_s, \log q_t)$ is taken to be jointly Gaussian with $\sigma_{\log q_t}^2 = \log(1 + \sigma_{q_t}^2 / (q_t^{(m)})^2)$, and $\vec{g}^{(m)}$ exponentiates the $\log q_t$ component. In either case the standard-Gaussian latent variable $\vec{\xi}$ is the variable of integration, so the Gauss-Hermite nodes and weights are unchanged. At each quadrature point $\vec{\xi}_{jk}$, the perturbed state $(h_s^{(m)}(\vec{\xi}_{jk}), q_t^{(m)}(\vec{\xi}_{jk})) = \vec{g}^{(m)}(\vec{\xi}_{jk})$ defines a local thermodynamic state through the saturation-consistent temperature $T^{(m)}(\vec{\xi}_{jk})$ recovered from Equation (??)

at fixed pressure and the prognostic-based liquid fraction

$$\lambda^{(m)} = \begin{cases} q_l^{(m)}/q_c^{(m)} & q_c^{(m)} > 0, \\ \mathcal{H}(T^{(m)} - T_{\text{freeze}}) & q_c^{(m)} = 0, \end{cases} \quad (1)$$

with $q_c^{(m)} = q_l^{(m)} + q_i^{(m)}$, T_{freeze} the bulk freezing temperature, and $\mathcal{H}$ the Heaviside step function, held fixed across all quadrature points of subdomain m . The Heaviside fall-back applies only when no prognostic condensate is present. Taking $\lambda^{(m)}$ from the prognostic phase composition rather than from a temperature-only prescription $\lambda(T)$ lets the SGS saturation adjustment inherit the subdomain's non-equilibrium phase memory: persistent supercooled liquid below T_{freeze} keeps $\lambda^{(m)}$ nonzero rather than reverting to zero at ice equilibrium. The saturation-adjustment derivatives in ?? use the same $\lambda^{(m)}$.

The saturation variable in the SGS coordinate is

$$S^{(m)}(\vec{\xi}) = \begin{cases} q_t^{(m)}(\vec{\xi}) - q_{v,\text{sat}}(T^{(m)}(\vec{\xi})) & \text{Gaussian SGS,} \\ \log\left(\frac{q_t^{(m)}(\vec{\xi})}{q_{v,\text{sat}}(T^{(m)}(\vec{\xi}))}\right) & \text{Lognormal SGS,} \end{cases} \quad (2)$$

and the local equilibrium cloud condensate at the same perturbed state is

$$q_l^{*,\text{eq}}(\vec{\xi}) = \max\left(0, \lambda^{(m)}[q_t^{(m)}(\vec{\xi}) - q_{v,\text{sat}}(T^{(m)}(\vec{\xi}))] - q_r^{(m)}\right), \quad (3)$$

with the symmetric expression for $q_i^{*,\text{eq}}(\vec{\xi})$ using $(1 - \lambda^{(m)})$ and $q_s^{(m)}$. The grid-mean precipitation $(q_r^{(m)}, q_s^{(m)})$ is subtracted from the equilibrium cloud partition so that $q_l^{*,\text{eq}} + q_i^{*,\text{eq}} + q_r^{(m)} + q_s^{(m)}$ never exceeds the local saturation excess. A single Gauss-Hermite quadrature over the $\vec{\xi}$ PDF then yields the cached moments

$$\mu_S^{(m)} = \langle S^{(m)} \rangle, \quad \sigma_S^{2,(m)} = \langle (S^{(m)})^2 \rangle - (\mu_S^{(m)})^2, \quad M_l^{(m)} = \langle q_l^{*,\text{eq}} \rangle, \quad M_i^{(m)} = \langle q_i^{*,\text{eq}} \rangle, \quad (4)$$

which require no microphysics calls and are used by both the cloud-fraction and the microphysics closures below.

Cloud fraction. We approximate the cloudy probability mass by a variance-matched logistic CDF in the coordinate of the saturation variable $S^{(m)}$,

$$f_c^{(m)} \approx \frac{1}{2} \left[1 + \text{erf}\left(c_f \frac{Q_{\text{eff}}^{(m)}}{\sqrt{2} \sigma_{q_c}^{(m)}}\right) \right], \quad (5)$$

with effective excess

$$Q_{\text{eff}}^{(m)} = \begin{cases} q_c^{(m)} + \min(0, \mu_S^{(m)}) & \text{Gaussian SGS,} \\ \log(1 + q_c^{(m)}/q_{v,\text{sat}}^{(m)}) + \min(0, \mu_S^{(m)}) & \text{Lognormal SGS,} \end{cases} \quad (6)$$

where $q_{v,\text{sat}}^{(m)} = q_{v,\text{sat}}(T^{(m)})$ is the saturation specific humidity at the subdomain mean state and $q_c^{(m)} = q_l^{(m)} + q_i^{(m)}$ is the prognostic mean condensate. The activation-scaled standard deviation is

$$\sigma_{q_c}^{(m)} = \alpha_c^{(m)} \sigma_S^{(m)}, \quad \alpha_c^{(m)} = \text{clamp} \left(\frac{q_c^{(m)}}{\sqrt{(q_t^{(m)} - q_{v,\text{sat}}^{(m)})^2 + \epsilon_q^2}}, 0, 1 \right), \quad (7)$$

with ϵ_q a small smoothing scale on the specific humidity and $(\cdot)_+ = \max(0, \cdot)$. The activation factor $\alpha_c^{(m)}$ caps the SGS spread used in the cloud-fraction transition by the amount of cloud water the prognostic state can support relative to the equilibrium-saturated excess at the mean; as $q_c^{(m)} \rightarrow 0$ in a clear subsaturated subdomain, $\alpha_c^{(m)} \rightarrow 0$ and $\sigma_{q_c}^{(m)} \rightarrow 0$, driving $f_c^{(m)} \rightarrow 0$ smoothly. The non-equilibrium offset enters through the numerator $Q_{\text{eff}}^{(m)}$, which recovers the prognostic condensate $q_c^{(m)}$ (or its log analog) in saturated regimes while forcing $f_c^{(m)} \rightarrow 0$ in deeply subsaturated regimes even when a numerical trace of prognostic condensate remains. The shape-preserving fitting factor c_f defaults to 1, recovering the form of the Sommeria-Deardorff cloud-fraction formula [?, ?, ?]; it absorbs higher-order corrections from approximating the SGS PDF.

Taking $\sigma_S^{2,(m)}$ and $\mu_S^{(m)}$ directly from the quadrature (4) ensures that the cloud fraction calculation shares the same SGS PDF that the microphysics integrates over. For lognormal SGS, the log coordinate handles the heavy upper tail of q_t through the variable of integration. The cloud-fraction equation (5) is solved together with the variance and mixing-length closures (which depend on $f_c^{(m)}$) as a self-consistent Picard fixed point at each time step.

Microphysics tendencies via the shape function. Microphysical tendencies are evaluated as the SGS-mean of the bulk microphysical rates, sampled on the same Gauss-Hermite grid that produced the moments (4). At each quadrature point $\vec{\xi}_{jk}$, we reconstruct the local cloud condensate from the prognostic mean through a shape function that distributes $q_\mu^{(m)}$ across the SGS PDF in proportion to the local equilibrium cloud water:

$$q_\mu^{\text{local}}(\vec{\xi}_{jk}) = q_\mu^{(m)} [\gamma_\mu^{(m)} q_\mu^{*,\text{eq}}(\vec{\xi}_{jk}) + \beta_\mu^{(m)}], \quad \mu \in \{l, i\}, \quad (8)$$

with blending coefficients

$$\gamma_\mu^{(m)} = \frac{M_\mu^{(m)}}{(M_*^{(m)})^2 + (M_\mu^{(m)})^2}, \quad \beta_\mu^{(m)} = \frac{(M_*^{(m)})^2}{(M_*^{(m)})^2 + (M_\mu^{(m)})^2}, \quad (9)$$

and a small blending scale $M_*^{(m)} \sim q_{\min}$ on the specific humidity. The coefficients satisfy $\gamma_\mu^{(m)} M_\mu^{(m)} + \beta_\mu^{(m)} = 1$ by construction, so that the SGS-mean of (8) is

$$\langle q_\mu^{\text{local}} \rangle = q_\mu^{(m)} [\gamma_\mu^{(m)} M_\mu^{(m)} + \beta_\mu^{(m)}] = q_\mu^{(m)}. \quad (10)$$

The reconstruction (8)–(9) therefore preserves the prognostic cloud-water and cloud-ice mass under the SGS-mean operator exactly, independent of the quadrature order. In the typical regime where the SGS PDF contains a saturated band (a cloud) so that $M_\mu^{(m)} \gg M_*^{(m)}$, $\beta_\mu^{(m)} \approx 0$ and $\gamma_\mu^{(m)} \approx 1/M_\mu^{(m)}$, it recovers the shape function

$$q_\mu^{\text{local}}(\vec{\xi}) \approx \frac{q_\mu^{(m)}}{M_\mu^{(m)}} q_\mu^{*,\text{eq}}(\vec{\xi}),$$

which concentrates the prognostic condensate in the saturated portions of the SGS PDF where it physically belongs. In the stale-cloud edge case where the saturated band has just vanished from the SGS PDF—a remnant of past condensation that has drifted into a now-subsaturated cell— $M_\mu^{(m)} \rightarrow 0$ and $\beta_\mu^{(m)} \rightarrow 1$, smoothly reverting to a uniform fallback $q_\mu^{\text{local}} \rightarrow q_\mu^{(m)}$ at every quadrature point that evaporates against the local subsaturated vapor.

The local thermodynamic state at quadrature point (j, k) is then

$$(T^{(m)}(\vec{\xi}_{jk}), q_t^{(m)}(\vec{\xi}_{jk}), (q_l^{\text{local}})_{jk}, (q_i^{\text{local}})_{jk}, q_r^{(m)}, q_s^{(m)}),$$

with vapor recovered as the residual $(q_v^{\text{local}})_{jk} = q_t^{(m)}(\vec{\xi}_{jk}) - (q_l^{\text{local}})_{jk} - (q_i^{\text{local}})_{jk} - q_r^{(m)} - q_s^{(m)}$, floored at zero. Bulk microphysical rates $\dot{Q}_{jk}$ are evaluated at this state and aggregated as

$$\overline{\dot{Q}}^{(m)} = \frac{1}{\pi} \sum_{j=1}^{n_j} \sum_{k=1}^{n_k} \omega_j \omega_k \dot{Q}_{jk}, \quad (11)$$

with $\{\omega_j, \omega_k\}$ the Gauss-Hermite weights. The shape-function reconstruction unifies the cloudy and clear contributions to the SGS-mean tendency in a single quadrature pass: subsaturated quadrature points where $q_\mu^{*,\text{eq}}$ vanishes carry only the small uniform fallback $\beta_\mu^{(m)} q_\mu^{(m)}$, which evaporates against the subsaturated local vapor and captures below-cloud rain evaporation and snow sublimation; saturated points carry the bulk of the cloud condensate and drive autoconversion, accretion, and condensational growth at rates that scale with $(q_\mu^{\text{local}})_{jk}$. Each component of $\dot{Q}_{jk}$ (the rates for q_l , q_i , and the precipitation categories) is aggregated independently by (11); total water and total energy are preserved at the subdomain level because the bulk microphysical rates preserve them at each quadrature point and the aggregation is linear. We use $n_j = n_k = 3$ Gauss-Hermite points by default, following Lopez-Gomez22a; this resolves the leading microphysical moments to within a few percent of higher-order quadrature for typical SGS variances.

2 My Review

- If I understand correctly, this method happily oversubscribes supersaturated areas if we ask for more condensate than there is saturation excess, since $\gamma \rightarrow 1$ and then we end up with just scaling the local condensate by the local over the cached value? Did we forget to apply the clamping that caps deviations? Last time I thought we worked in deviation space so that things got conserved properly, even for cloud fraction you apply α_c scaling.
- Handing off to a uniform distribution as a default still strikes me as a poor idea. As a matter of principle it's overly aggressive. Why do this?
- The shape parameters however are not bad, they do hand off fairly smoothly as far as I can tell
- The shape parameters I guess still can be modified to account for physical processes, e.g. slow microphysics etc.
- The cloud fraction seems odd to me. What exactly is c_f ? where does it come from? If i understand correctly, subsaturated clouds only count once they're big enough to evaporate to saturation? why? If the argument is those are the only clouds that would exist under instantaneous adjustment, then you could make the argument for using $\max(\mu_s, q_c)$ also. The very existence of these nonequilibrium clouds should imply that fast adjustment relative to the generating process does not happen, so it feels weird to make this dichotomy and not just rely on the actual condensate. Is this just for downstream regimes that need to assume moist adiabat assumptions? It still feels like we'd be better off not mixing the cloud to uniformity and actually using something better in that case.

You write: $q_c = \bar{q}_c + \bar{q}_c \gamma_l (q_c^* - q_c^*)$. But really you need to ensure that $\bar{q}_c \gamma_l (q_c^* - q_c^*) \approx \bar{q}_c \frac{1}{q_c^*} (q_c^* - q_c^*)$ doesn't exceed the true saturation span q_c^* , that is deviations are never larger than the original. So we need to ensure: $\bar{q}_c \gamma (q_c^* - q_c^*) < (q_c^* - q_c^*) \Rightarrow \gamma < 1/\bar{q}_c$, so $\gamma = \min(\gamma_0, 1/\bar{q}_c)$, which would mean we'd need to adjust β as well, $\left(\frac{1}{q_c}\right) M + \beta = 1 \implies \beta = 1 - \frac{M}{q_c}$ if clamped.

Even if you argue that you'd rather have the condensate stay in the cloudy portion, you risk severe shocks going from having even a small amount of cloud (with all the condensate piled in there) to no cloud uniform distribution... Just seems like a bad move. There's not a large difference between say full 1% subsaturation at the highest grid point with $q_c = 1e-5$ and 1 quadrature point barely saturated, if say $q_c^* > M_*$, but is not large like e.g. $1e-8$, but these get treated vastly differently. The former gets a uniform spread, the latter piles everything onto the saturated points. (need to double check this goes gracefully after clamping, and doesn't create any weird fluctuations transitioning between the subsat and saturated part of the grid - right now i think it actually will dip down and up again bc the supersaturated region perturbations do that, and β will no longer be too small for you to see it.

3 My Frameworks

Mass-Conserving Continuous Variational Scheme

The truncated additive scheme successfully conserves mass but creates unphysical, uniform plateaus of condensate in subsaturated regions because the truncation $q^* = 0$ destroys thermodynamic gradient information outside the cloud. To maximize the saturation excess that subgrid condensate is exposed to—while naturally tapering out in drier downdrafts—we frame the subgrid distribution as a constrained variational problem (a “water-filling” algorithm).

TODO: Reformulate so we have fluctuations pre-scaled with latent heat into Δq space instead of ΔS , I used to use $\Delta q_c \approx \Delta S/\Gamma_c$, but one can imagine errors from this setup. Unlike the original formulation, we should not stop at 0, so I wonder if it’s possible to saturation adjust with negative condensate to get q_c or do a first order linearized estimation Δq_c from ΔS

We start in the continuous limit. Given the saturation excess $S = q_t - q_s^*$, we define $S' = S - \bar{S}$ (we could also work in normalized units like $q'_c - \bar{q}_c$ to alleviate some latent heating calculations). We define the local response capacity $\alpha S'$, where α is some parameter that defines how strongly condensate responds to SGS fluctuations, and infer the full potential:

$$P = \lambda_0 + (\bar{S} + \alpha S') = \lambda + \alpha S' \quad (12)$$

i.e. the local capacity to contain condensate where λ_0 is some offset that encodes nonequilibrium memory. So in the equilibrium framework, λ_0 is 0, and α is 1.

Let $q_{c,\alpha}^*$ be the saturation equilibrium condensate, that is, the amount of condensate in saturation equilibrium, accounting for the assumed α response to fluctuations. Then, our solution falls into two regimes.

First, $q_c \leq q_{c,\alpha}^*$. In this case, we proportionally scale condensates down by $q_c/q_{c,\alpha}^*$. We prefer this scaling over shifting λ since shifting would result in a large part of the supersaturated fraction of the domain going without condensate. In some other cases it would be possible to simply scale α , but this is not always possible as the mean of the distribution scaling asymptotes to might exceed our requested q_c so it is not a reliable method, and can over-damp local variability.

This does add a branch to the algorithm though [TODO :: look up how ClimaAtmos handles this pattern right now, surely there’s a lot of places where it’s worthwhile to skip expensive microphysics for example)]...

Second, if $q_{c,\alpha}^* < q_c$, then we just aim to shift λ . We can note then that λ will always be positive, but solving this equation requires root-finding. Section **TBD** discusses some methods for optimizing root-finding performance.

From a theory perspective, we believe this branching structure to be optimal. When condensate is less than expected, we can expect that in the SGS we have different regions growing and approaching their individual asymptotes. This is safer than assuming the condensate came from elsewhere, and better than trying to impose full size fluctuations on top of the real data. While

it's possible a shift or scaling in α could work, there is no way to know which option is certain and scaling down saturation condensate is analytically possible, fast, and ensures all saturated bins are treated fairly without necessary further assumptions. When condensate is more than expected, we can say for certain it came from somewhere else, meaning it has had a longer time to homogenize, and then the fluctuations in its structure are imposed by the evaporative process.

- Plot distributions from real data. We have GoogleLES which is saturation adjusted though, we could pull some other LES (are those arctic clouds around?) though for rn we really only are set up to compare to single condensate data.

TODO: think if there's an algorithm that transitions neatly from shifting to scaling to reduce excess condensate. Is it possible for regimes to blend nicely without untoward side effects?

Note this framework does permit using α values that are not equal to 1 in *any* scenario; for example one might assume some smaller value to represent a slower microphysical response time to saturation fluctuations, or some larger value at coarser resolutions when SGS variability is more structural than fluctuation, or calibrate against GCMs – using $\alpha=1$ just maps nicely to saturation adjustment assumptions.

So, if $q_c^{(m)} > q_c^{*(m)}$ we can turn to a root-finding approach to optimize λ , where

$$q_c^{(m)}(\lambda) = \int dS' \max(0, \lambda + \alpha S') = q_c^{(m)} = \frac{\alpha \sigma_S}{\sqrt{2\pi}} \exp\left(-\frac{\lambda^2}{2\alpha^2 \sigma_S^2}\right) + \lambda \left[\frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\lambda}{\alpha \sigma_S \sqrt{2}}\right) \right] \quad (13)$$

and we solve $\mathcal{R}(\lambda) = q_c^{(m)} - q_c^{*(m)}(\lambda) \rightarrow 0$.

Given λ , we can trivially calculate the cloud fraction as the integral of the PDF over all regions where condensate exists (i.e., where $\lambda + \alpha s' > 0$, or $s' > -\lambda/\alpha$):

$$\text{CF}(\lambda) = \int_{-\lambda/\alpha}^{\infty} \mathcal{P}(s') ds' = \frac{1}{2} + \frac{1}{2} \operatorname{erf}\left(\frac{\lambda}{\alpha \sigma_S \sqrt{2}}\right) \quad (14)$$

For model purposes, note that this may not be the quantity you wish to use in every scenario. For example, if you are assuming fast responses via latent heating to remain on an adiabat, this strictly speaking only tells you where you have allocated condensate and have capacity for such a response (i.e. probably valid for liquid, but a poor approximation for ice where the response is slow, and in subsaturated regions with condensate, the cloud fraction will return 1).

3.1 Discrete Form

The steps from the continuous distribution can be mapped easily to the quadrature points if exact conservation is required (though this requires repeated evaluations over the quadrature points), but if going directly from the continuous

form, the condensate value at any single quadrature point can be calculated without cache. The strictly discrete formulation is written as follows:

We utilize the continuous, zero-mean thermodynamic fluctuation of the raw saturation excess, $S_k = q_{t,k} - q_{s,k}^*$:

$$\Delta S_k = S_k - \sum_{j=1}^{N_q} w_j S_j, \quad q_{\mu,k}^{local} = \max\left(0, \lambda + \alpha_\mu^{(m)} \Delta S_k\right) \quad (15)$$

$$\mathcal{R}(\lambda) = \sum_{k=1}^{N_q} w_k \max\left(0, \lambda + \alpha_\mu^{(m)} \Delta S_k\right) - q_\mu^{(m)} \rightarrow 0 \quad (16)$$

It is possible to enforce exact conservation under this framework, but requires N^2 evaluations at each λ during root-finding. **Actually under first order linearization you can just do N^2 once at beginning and then assume shifts are equal and maybe apply a corrective factor at the end to be certain, so maybe $2N^2$ evaluations.**

Whether or not to pursue exact conservation of q_c across quadrature points, or pointwise analytical exactness is left to the users chosen implementation. We opt for speed and correctness to not enforce bulk q_c convergence over quadrature for several reasons, first the correction can be 10-20% at reasonable values like $N=5$ which, given we do no such bulk sum enforcement for thermodynamic or total liquid, can create spurious saturation deviations. Second, doing so imposes additional computational cost for little gain.

3.1.1 Controlling homogeneity (adapting the variance scalar α)

The baseline potential assumes condensation is infinitely fast and tracks local subgrid fluctuations perfectly. In reality, the spatial distribution of condensate is determined by a competition between turbulent mixing (which smears the distribution uniformly) and microphysical condensation (which sharpens the distribution into the wettest subgrid pockets). This formulation provides a physically robust transition between two limits:

- **Fast Microphysics** ($\alpha_x \rightarrow 1$): The potential feels the full amplitude of the thermodynamic perturbations. The solver forces λ negative to constrain the mass, sharply confining condensate to the highest supersaturation peaks (the “water-filling” limit).
- **Well-Mixed** ($\alpha_x \rightarrow 0$): Turbulence dominates, destroying local subgrid perturbations. The equation collapses to $P_x = \max(0, \lambda + \bar{S})$, and every q_c becomes uniform across the distribution.

We discussed how $\alpha=1$ is the most aggressive form of this implementation. In principle, in real clouds, we are often not so strictly in saturation adjustment, or may have other reasons to not wish to be so aggressive. We could consider piloting, rather than fixed α values, functional α equations that adapt to local

dynamics. For example here we imagine some homogenization factor based on the Damköhler number:

$$\alpha_{\text{hom}} = k \frac{\tau_{\text{mix}}}{\tau_{\text{mix}} + \tau_{\text{micro}}} (\text{or anything else}), 0 \leq \alpha \lesssim 1 \quad (17)$$

adapted from and very much in line with arguments made about the shape of the supersaturation distribution being driven by turbulence, mixing, etc...

(there are a few papers about setting the SGS supersaturation variance for e.g. mixed-phased clouds but these I think would struggle to generalize to GCM scales and spatial heterogeneity etc, so something rough I think works out) We have not implemented any such framework at the moment, but consider this a direction worth pursuing. For now we just take as a calibratable parameter.

Note that we did consider alternate frameworks that smoothly transition from a fluctuation-driven framework and a well-mixed uniform framework by using α not as a fluctuation response but as a linear mixing factor:

$$q_c = \alpha_{\text{hom}} q_{c,\text{WF}} + (1 - \alpha_{\text{hom}}) q_c^{(m)} \quad (18)$$

However we consider such frameworks to be not ideal as we do not think it is ever truly ideal to enforce homogenization, as this leads to unnecessary condensate depletion that is not easily calibrated out. The α as a fluctuation response is very adaptable, has no sudden transition to homogeneity, and is readily adjusted to local dynamics, model resolution, or other factors. For example, in a spatially similar stratocumulus context one might use α to consider mixing vs microphysics timescales, but when modeling inhomogeneous convection at 100km resolution, condensate is likely truly highly correlated with moisture, and one might adopt something much closer to $\alpha=1$.

Fast root solving methods

Operationally, the equation to solve for λ from Equation 13 can be a little bit stiff. This is primarily because while the main part of the distribution is well behaved, in the tails the decay of the fluctuations becomes exponential, making combining linear shifts with this exponential tail challenging for initial optimization and fast solves. Naive bracketing, Newton-Raphson, and other methods routinely achieve 1% accuracy within 5-8 iterations. However, for online model performance, we require fast, single iteration, special-function free, branchless methods.

We start with a few simple approximations for the erf and CDF of the Gaussian from Winitzki [\[need citation\]](#)

$$\text{erf}(x) \approx \sqrt{1 - \exp\left(-x^2 \frac{\frac{4}{\pi} + ax^2}{1 + ax^2}\right)}, \quad \Phi(x) \approx \frac{1}{2} + \frac{1}{2} \sqrt{1 - \exp\left(-\frac{x^2}{2} \frac{\frac{8}{\pi} + \frac{a}{2}x^2}{1 + \frac{a}{2}x^2}\right)} \quad (19)$$

I think we found for values above 3 this starts to lose accuracy (as do steps based on it)? it might pay to switch to a log-asymptotic version there? But that adds a branch...

These are fast, accurate, and importantly, invertible.

Solvers built with this method to date benchmarked (on sampo) at ~230ns per root-solve on CPU (no GPU special function units).

Operationally, we believe that is too slow. Additionally it remains difficult to transition between the central linear region and the tails. Particularly in dry tails (where q_c exists but is far in the tail of the distribution for the required shift), things are particularly challenging. [\[Note that with \$N^2\$ quadrature points with pointwise, at some point these tail values hardly show up at all, and by definition this usually corresponds to fairly small condensate amounts, but it's not trivial, I struggled to represent \$q_c = 10^{-6}\$ from inputs \$\rho=1, T=280, \mu = -3 \times 10^{-3}, q_c=10^{-6}, q_t=q^*+\mu, \sigma_q=\(10^{-10}\), \sigma_T^2=0.25, r_{Tq}=0, \text{LF}=0.4\$](#)

We thus further optimized a faster solver that uses an offline optimized rational function approximation that smoothly blends between a ‘moist’ and a ‘dry’ limit, with a single log-space Newton step refinement. Solvers built with this method to date benchmarked (on sampo) at ~67ns per root-solve on CPU.

Details are in [ClimaAtmos.jl::jb/sgs-quadrature](#) (the fast version has not been operationalized yet as of 2026/05/17)

3.2 Sample 1D results

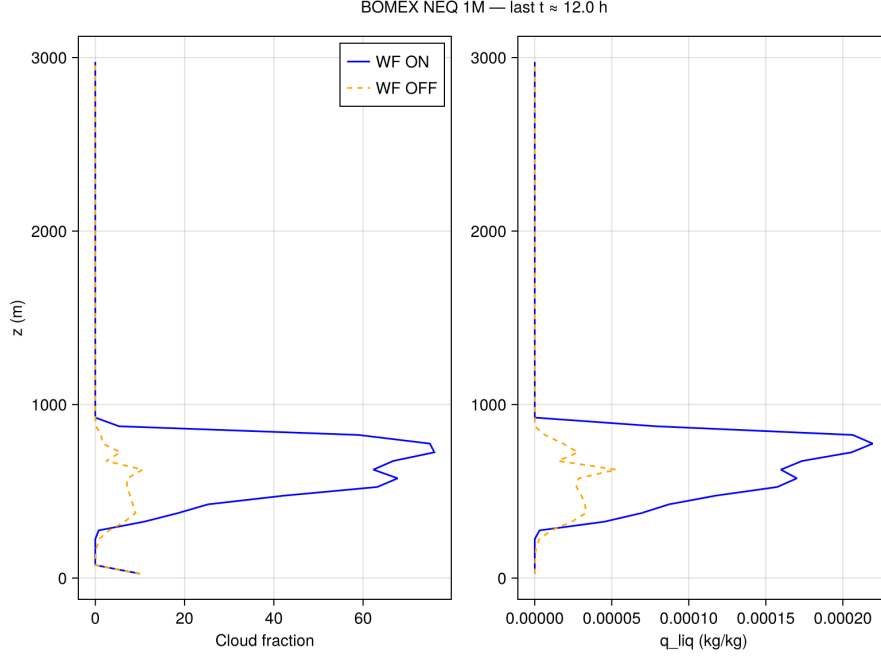


Figure 1: BOMEX comparing uniformly distributed condensate $\alpha=0$, to full strength water-filling method $\alpha=1$, after 12 hours of BOMEX simulation. Note that this is not a checkpoint redistribution, for that see Figure 2, these models diverge due to differing TKE evolution and other factors that make the α response nonlinear to chaotic. However, we do see markedly better condensate preservation. These are uncalibrated, so performance could be better, but relative to it looks like we’re still under-producing condensate by about a factor of 4, but this is a step in the right direction. Relative to FastEddy it also looks like α that decays towards cloud top might be ideal, which potentially makes sense given their TKE and σ_w^2 increase near cloud top and ours does not fully, but further reasoning and inspection was not done.

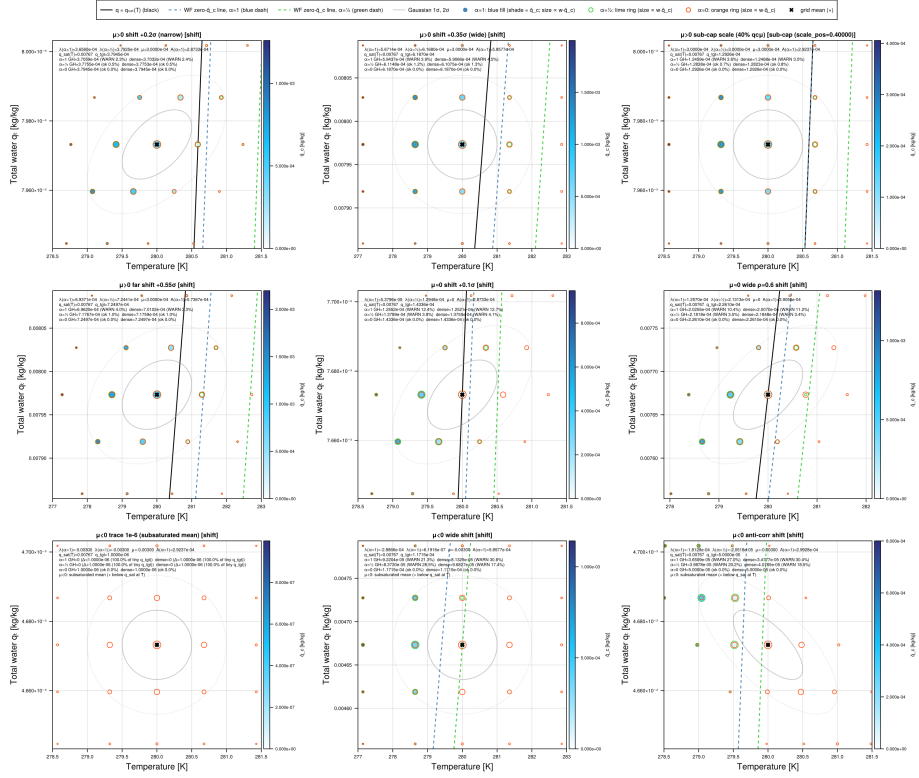
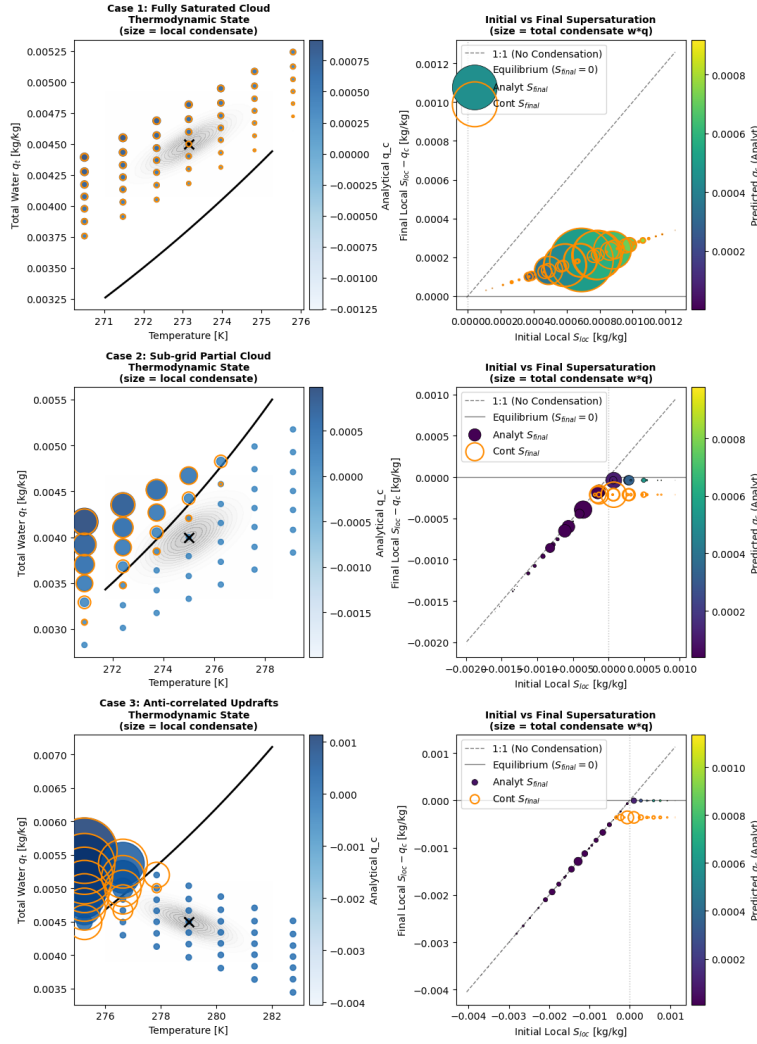


Figure 2: Sample $N=5$ quadrature points with $\alpha = 0$ (orange rings), 0.5 (green rings), and 1 (blue circles). The water filling shifted boundary is shown in the same colors as the points for $\alpha=0.5, 1$ as well. Many of these test cases were chosen to be challenging (note this is still not the rational approx 67 ns solver which performs better, I haven't regenerated with those yet [2026/05/17])



```

--- INPUT PARAMETERS ---
T_mean: 273.15 K
qt_mean: 4.50e-03 kg/kg
Input <q>: 5.00e-04 kg/kg
Eq <q>: 6.850e-04 kg/kg

```

--- CONDENSATE PARTITIONING TABLE ---					
Scheme	Total <q>	Super <q>	Sub <q>	CF	Wgt-S_f
Analyt	5.000e-04	5.000e-04	1.602e-16	1.000	1.91e-04
Cont	5.000e-04	5.000e-04	0.000e+00	1.000	1.91e-04

```

--- INPUT PARAMETERS ---
T_mean: 275.00 K
qt_mean: 4.00e-03 kg/kg
Input <q>: 5.00e-05 kg/kg
Eq <q>: 1.135e-05 kg/kg

```

--- CONDENSATE PARTITIONING TABLE ---					
Scheme	Total <q>	Super <q>	Sub <q>	CF	Wgt-S_f
Analyt	5.000e-05	1.532e-05	3.468e-05	1.000	-3.32e-04
Cont	5.000e-05	3.321e-05	1.679e-05	0.337	-2.13e-04

```

--- INPUT PARAMETERS ---
T_mean: 279.00 K
qt_mean: 4.50e-03 kg/kg
Input <q>: 1.00e-05 kg/kg
Eq <q>: 1.343e-06 kg/kg

```

--- CONDENSATE PARTITIONING TABLE ---					
Scheme	Total <q>	Super <q>	Sub <q>	CF	Wgt-S_f
Analyt	1.000e-05	1.420e-06	8.580e-06	1.000	-1.13e-03
Cont	1.000e-05	4.564e-06	5.436e-06	0.038	-3.62e-04

Figure 3: WF test (old example)

Separating Phases: Relational Potential Partitioning (RPP) (not implemting yet)

While our 1D closure works with assumed liquid fixed liquid fractions, liquid fraction is decidedly not fixed at the sub-gridscale. There are many reasons for this. SGS variability spans subsaturation, WBF regimes, and supersaturation differently for liquid and ice, and liquid responds very quickly to supersaturation fluctuations while ice often does not. Thus we seek to determine a fast, 2D partitioning plan that jointly allocates liquid and ice q_l and q_i at the SGS. This is a much harder problem, particularly because there is not a single clear answer – for example while liquid may responds fastest to supersaturation fluctuations, that does not mean that we can give it priority in consuming saturation, at long time horizons ice will grow to compete with liquid.

Thus, the goal of the Relational Potential Partitioning (RPP) scheme is to diagnostically reconstruct the continuous subgrid spatial distributions of liquid (q_l) and ice (q_i) such that they strictly conserve the grid-mean targets, while physically accounting for local thermodynamic capacities, turbulent mixing timescales, and phase competition.

We define two separate 1D potentials for liquid and ice

$$P_l(\lambda_l) = \lambda_l + (\overline{S}_l + \alpha_l S') \quad P_i(\lambda_i) = \lambda_i + (\overline{S}_i + \alpha_i S') \quad (20)$$

where again we take the fluctuations in supersaturation to be equal for both condensates. Then, the shared-envelope total condensate is $q_c = \max(P_l(\lambda_l), P_i(\lambda_i))$, which is proportioned into phases as $q_x = q_{tot}(P_x/(P_l + P_i + \epsilon))$. The task then becomes to solve for λ_l and λ_i

We make one final modification. Recent literature [\[TODO: add here\]](#) has strongly indicated that liquid and ice overlap at the SGS can not be assumed to be uniform. To incorporate this effect, we introduce Ω , an overlap parameter that quantifies how strongly liquid and ice interact. When $\Omega \rightarrow 0$, liquid and ice do not overlap at all, and instead of a joint system we solve liquid and ice completely separately, $q_x = \max(0, P_x)$. [Note here we haven't worked in how to apply the \$q_x\$ less than saturation \$q_x\$ scaling option like we did before, I need to think more about how this works logistically. Maybe it's applied to \$q_c\$ still? idk. And I need to add test cases that cover this. For the first pass I just didnt worry about it I think \(verify python code\)](#)

Note this does create some deficiencies, for example we do not repeat our N^2 quadrature separately for liquid and ice so the results of the partitioning do get recombined (this is actually pretty bad and I need to think of a way to fix this... I think the easiest way would be to pass Ω into the micro-physics solvers so that liquid and ice are treated with separate q_t , T , etc. In `TurbulenceConvection.jl` SOCRATES experiments we didn't quite go so far, but we did permit setting a mixed-phase cloud fraction that scaled the strength of the WBF $(q_{sl} - q_{si})/\Gamma_i \tau_i$ term in the MM2015 framework. Ideally I think we would separate $T_{l,i}$, $q_{t(l,i)}$ and define $q_{l(i)}$ and $q_{i(l)}$ when necessary for tendencies that need them (extend to snow and rain accordingly), or splitting into an all

liquid, all ice, and mixed-phase contribution. In the other limit, $\Omega \rightarrow 1$, we solve the full RPP equation.

Our best method for utilizing this overlap parameter is to linearly blend between the two limits **[TODO :: think, can we do better?]**:

$$q_\ell(\psi) = (1 - \Omega)P_\ell + \Omega \left[Q_{\text{tot}} \left(\frac{P_\ell}{P_\ell + P_i + \epsilon} \right) \right] \quad (21)$$

$$q_i(\psi) = (1 - \Omega)P_i + \Omega \left[Q_{\text{tot}} \left(\frac{P_i}{P_\ell + P_i + \epsilon} \right) \right] \quad (22)$$

where $\psi = (q_t, T, \dots)$ describes the local model state e.g. at any quadrature point.

3.3 System Closure

The subgrid distribution is fully resolved by enforcing the host model's grid-mean mass targets as hard physical constraints. We integrate the unified point-wise equations across the subgrid probability density function (PDF) (or spatial quadrature weights $w(\psi)$):

$$q_\ell^{(m)} = \int d\psi q_\ell(\psi, \lambda_\ell, \lambda_i) \quad (23)$$

$$q_i^{(m)} = \int d\psi q_i(\psi, \lambda_\ell, \lambda_i) \quad (24)$$

This forms a closed system of two non-linear equations with exactly two unknowns (λ_ℓ and λ_i). A standard root-finding algorithm determines the exact spatial shifts required to conserve mass, automatically adjusting the phase locations to continuously respect the turbulence bounds (α) and the physical volume exclusion (Ω). Once the shifts are found, one readily obtains cloud fractions (including the shared overlap for WBF, $\text{CF}_{\text{shared}} = \Omega \min(\text{CF}_\ell, \text{CF}_i)$), where we can note that we can estimate the cloud fraction without actually considering the full λ_i, λ_ℓ relation (at least if we assume having any condensate means cloud). If you did do something like set a minimum condensate threshold for considering a cloud, I think this still holds up, it only gets hard if you set the threshold based on e.g. some minimum amount of liquid only as it's the fast condensate or something like that, otherwise I think things are pretty straightforward.

Actually finding the λ values in 2D space is, however, quite challenging to make numerically stable due to stiff equations (see the loss landscape plots in Figure 4)

3.4 Fast 2D root solves

Not Implemented Yet

3.5 Sample 2D (RPP) results

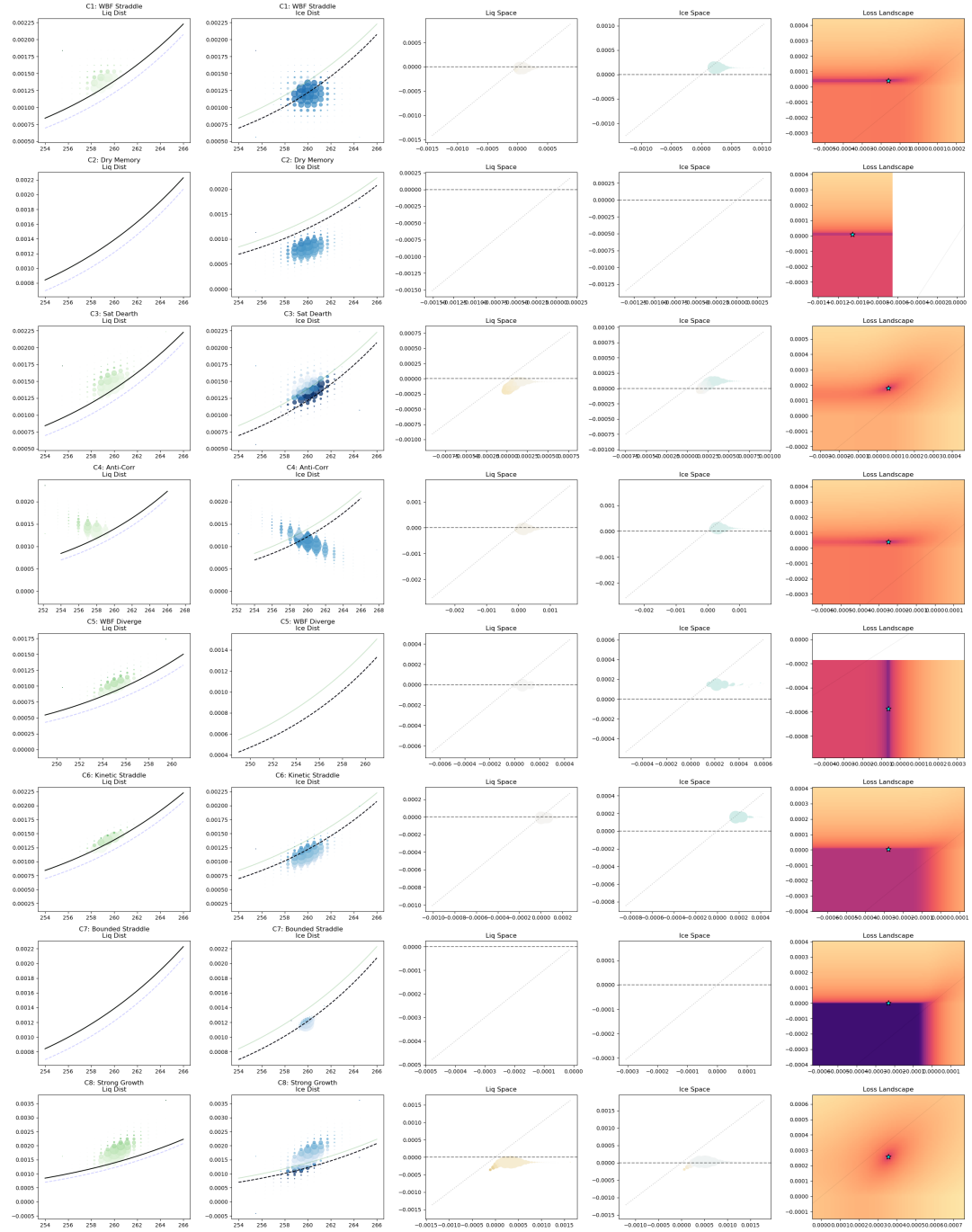


Figure 4: Various tests of the RPP framework, run with $\alpha_l=1$, $\alpha_i=0.01$, and $\Omega=1$ at $N=7$. We can see that while the method is not perfect, it has several strengths.. For example we can readily see WBF ice enhancement and stronger liquid response to local saturation. We do note that the constraint framework has its limits, for example in C8 (and in some other places) we see q_i peaking below ice saturation (with low α_i it's important to note that all these points experience roughly $\bar{S}_i$ so it's not strictly a failure), due to the joint liquid constraint, but we also see strong ice recovery in the jointly supersaturated area, and a stripe of low ice formation above the WBF regime, so it's not all bad. Critically however, note the stiff nature of the loss landscape **TODO :: Verify this is all coded up right, and add here what the loss function is**